

SMART-Scholar: A Scholarship Management and Tracking System

Ravigel T. Ablen¹, Shannyn C. Flores², Mark Vincen D. Pasturan³, and
Edward B. Bertulfo⁴

^{1, 2, 3, 4} Eastern Visayas State University – Ormoc Campus, Ormoc City, Leyte, Philippines
edward.bertulfo@evsu.edu.ph (Corresponding Email)

Abstract. The manual administration of scholarship programs in Ormoc City has long been plagued by inefficiencies, including delayed processing, inaccurate record-keeping, and poor communication among stakeholders. To address these challenges, this study introduces Smart-Scholar, an automated scholarship management and tracking system designed to streamline workflows, improve data integrity, and enhance communication between LGU coordinators, applicants, and scholars. Developed using the Agile methodology, Smart-Scholar incorporates iterative feedback into its development cycle, resulting in a responsive and user-centric platform. The system features dedicated web interfaces and a mobile application, enabling real-time updates, large-scale data management, and task automation. Evaluation through user testing and performance analysis revealed that 93% of participants rated the system as user-friendly and time-efficient, with substantial improvements in data accuracy and operational transparency. Minor concerns, such as occasional email verification delays and interface learning curves, were noted. Overall, the study concludes that Smart-Scholar effectively resolves the core issues of manual scholarship administration and demonstrates strong potential as a scalable, inclusive solution to support the educational advancement of Ormoc City's youth.

Keywords: Scholarship Management System, Agile Development, Data Accuracy, Educational Support, Ormoc City LGU Scholarship.

1. Introduction

The distribution and administration of scholarships are vital functions of local government units (LGUs) aimed at supporting the educational aspirations of deserving students, particularly those from economically disadvantaged backgrounds. In Ormoc City, the Local Government Unit has long managed its scholarship programs through conventional paper-based methods. While this manual system has sustained basic operations, it suffers from several critical inefficiencies that compromise service quality, transparency, and operational scalability.

Manual processing of scholarship applications results in delays during screening, verification, and approval, often causing backlogs and missed deadlines. The absence of centralized digital records complicates tracking of application status, scholar performance, and program compliance, hindering effective monitoring. Communication gaps between

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coordinators and scholars lead to missed updates, unclear requirements, and repetitive inquiries, increasing the administrative burden. Additionally, reliance on physical records raises data security risks due to potential loss or unauthorized access. The system's lack of remote access further limits participation for scholars in remote areas or studying outside the city.

In response to these identified problems, this study introduces SMART-Scholar, a web-based scholarship management and tracking system designed specifically for the local government of Ormoc City. Taking reference from the studies of Secugal *et al.* [1] and Al-Ayyubi and Maulana [2], the system aims to automate the end-to-end processes of scholarship administration while enhancing data integrity, transparency, and user accessibility. By leveraging cloud-based architecture, a secure database backend, and user-centric design principles, SMART-Scholar offers a scalable and sustainable solution tailored to the unique operational context of local government scholarship offices.

This study presents the design, development, and evaluation of SMART-Scholar using an Agile methodology and modern development tools such as Supabase, ReactJS, and Flutter. The effectiveness of the system is measured against both functional and non-functional criteria, demonstrating its capacity to resolve the previously identified inefficiencies and improve overall scholarship program management.

2. Related Literature

Several studies have explored the digitization of scholarship management systems, emphasizing the need for automation, improved communication, centralized data handling, and secure platforms to replace inefficient manual processes. The development of the SMART-Scholar system draws on concepts, techniques, and findings from these works to address the operational challenges faced by the Ormoc City LGU in managing its scholarship programs.

Sowndhariyaa and Nithya [3] proposed an advanced e-scholarship platform that utilizes content-based filtering techniques to match students with relevant scholarship opportunities. Their system was designed to overcome the limitations of traditional paper-based processes, which are often slow, error-prone, and administratively demanding. By enabling students to apply online, receive notifications, and track application statuses, their platform mirrors SMART-Scholar's core objective of streamlining scholarship processing and minimizing delays through end-to-end digital workflows.

One of the key contributions of their study is the use of automated filtering and recommendation systems to categorize applicants based on attributes such as caste, academic rank, and financial background. These techniques reduce administrative overhead during evaluations—an approach similarly adopted by SMART-Scholar through its applicant profiling and status monitoring tools. Furthermore, the incorporation of SMS and email notifications in their system to ensure timely communication with scholars reflects a parallel strategy in SMART-Scholar, which also employs notification systems to close communication gaps between stakeholders.

Additionally, the inclusion of varied scholarship categories (academic, skills-based, athletic, and co-curricular) in their model promotes inclusivity, another principle that

guided SMART-Scholar's development. These features collectively contribute to a personalized user experience, an aspect SMART-Scholar emphasizes through its role-specific interfaces and scholar dashboards.

In a related study, Komsari and Airlangga [4] developed a website-based scholarship information system for UNWAHA students. Their work responded to the lack of a centralized digital platform by introducing a user-friendly web interface to disseminate scholarship announcements and updates. This directly aligns with SMART-Scholar's goal of enhancing accessibility and engagement by providing a centralized, responsive portal where applicants and scholars can stay informed and connected.

Taken together, these related systems highlight the transformative potential of digital platforms in scholarship management. They emphasize key functionalities such as automated evaluations, real-time communication, centralized databases, and intuitive user interfaces—all of which are foundational to SMART-Scholar. The system builds upon these models by integrating modern technologies like Supabase for cloud database management, ReactJS and Flutter for scalable cross-platform access, and SSL encryption for data security. These implementations collectively ensure that SMART-Scholar meets the evolving needs of the Ormoc City LGU and its stakeholders.

3. Methodology

The SMART-Scholar system was developed using a structured, iterative approach grounded in software engineering principles and applied research. The study aimed to design, develop, and evaluate a scholarship management and tracking platform that addresses the inefficiencies of Ormoc City LGU's manual processes. A developmental research strategy was combined with Agile methodology to support continuous user feedback, flexible iterations, and progressive system refinement.

Requirements were initially gathered through informal interviews and process observation at the LGU scholarship office. This confirmed key challenges such as application delays, fragmented scholar data, limited communication, and inefficient reporting. These findings informed the system's functional and non-functional requirements.

Agile sprints guided development, with each cycle focused on core features including user authentication, profile and application management, notification services, and reporting tools. At the end of each sprint, prototype testing was conducted with end users to validate functionality and inform improvements.

SMART-Scholar follows a three-tier architecture composed of a presentation layer, application logic, and a backend database. The frontend was developed using ReactJS for web interfaces and Flutter for mobile accessibility. Supabase served as the backend platform, supporting PostgreSQL, real-time data handling, and secure authentication via API endpoints. This stack enabled a scalable, cloud-based solution with encrypted communication and reliable data access.

As illustrated in Figure 1, the system functions as a central hub for scholar-coordinator interactions. Scholars can apply, update personal data, and receive announcements; administrators manage user accounts and update records; LGU coordinators verify applications and access scholar profiles; and school coordinators

maintain school data and publish announcements. All records are stored in a centralized Scholar Database to ensure data consistency and traceability.

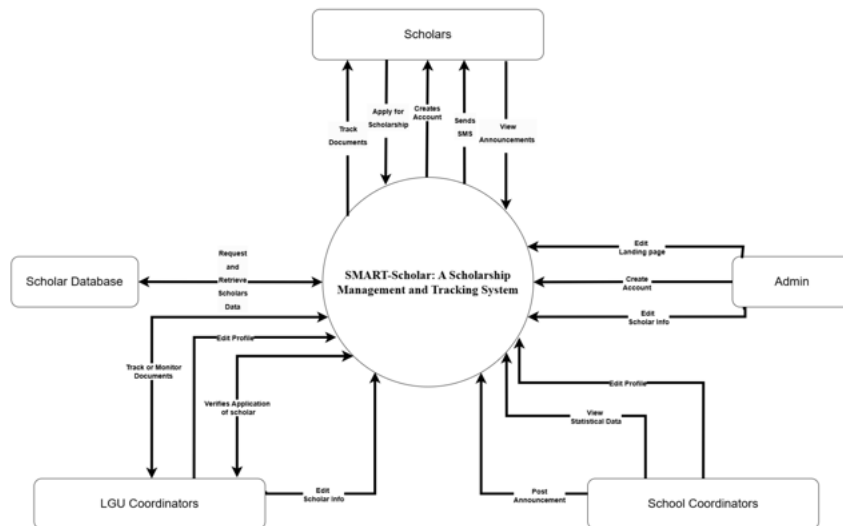


Figure 1. Data Flow Diagram of the System and Mobile Application

To define and validate system behavior, software models were created, including use case diagrams, data flow diagrams (DFDs), as shown in Figure 1, and entity-relationship diagrams (ERDs). These models mapped user interactions, data structures, and process logic across the platform.

Testing involved black-box evaluation of individual modules and full system integration. Usability testing gathered feedback from scholars and coordinators, while performance tests measured responsiveness and accessibility. Results confirmed that the system met functional requirements, maintained high reliability, and was well-received by stakeholders.

Overall, the applied methodology ensured the development of a technically sound and user-centric system that aligns with the operational goals of Ormoc City LGU and enhances the scholarship administration process.

4. Results and Discussion

The SMART-Scholar system underwent a comprehensive evaluation to assess functionality, usability, and its effectiveness in addressing identified operational challenges. Functional testing confirmed the reliability of key modules, including user registration, authentication, document uploads, application tracking, and notifications. No critical errors were encountered, and minor bugs were resolved through Agile-based iteration.

Non-functional testing focused on usability, performance, and security. While no formal Likert-scale metrics were used, both scholars and coordinators reported that the system was intuitive, responsive, and significantly improved over the manual approach. Scholars particularly appreciated real-time updates and document tracking, as shown

in Figure 2, while administrators observed enhanced data organization and faster processing.

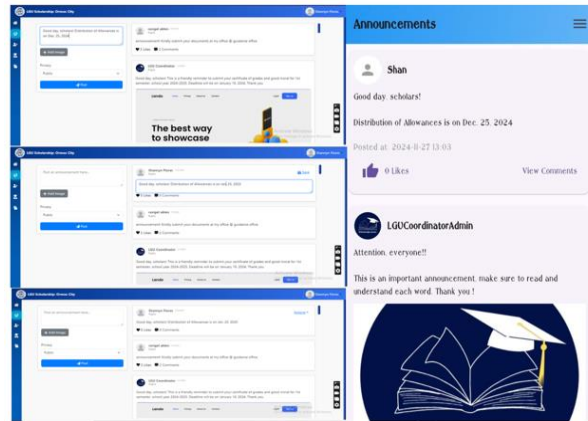


Figure 2. Announcement Feature Interface of SMART-Scholar

The system's backend, powered by Supabase, provided real-time data retrieval and consistent uptime, resolving earlier issues such as fragmented records and update delays. As shown in Figure 3, system uptime remained stable over a 90-day test period. Security was reinforced through SSL encryption to ensure the protection of sensitive scholar data.



Figure 3. Supabase Uptime During the 90-day Evaluation Period

5. Conclusion

This study presented the design, development, and evaluation of SMART-Scholar and addressed critical inefficiencies in manual scholarship administration by streamlining workflows, improving data integrity, and enhancing communication among stakeholders. Developed using Agile methodology and modern tools such as Supabase, ReactJS, and Flutter, SMART-Scholar demonstrated strong performance, usability, and accessibility during evaluation. Results showed significant improvements in administrative efficiency and user satisfaction. With its scalable architecture and successful implementation, SMART-Scholar offers a practical and adaptable solution for modernizing scholarship programs in other local government units and educational institutions.

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